

UNIVERSITY OF EDINBURGH
COLLEGE OF SCIENCE AND ENGINEERING
SCHOOL OF INFORMATICS

COMPUTER COMMUNICATIONS AND NETWORKS

Friday 27th August 2010

09:30 to 11:30

Year 3 Courses

Convener: K. Kalorloti
External Examiners: K. Eder, A. Frisch

INSTRUCTIONS TO CANDIDATES

Answer any TWO questions.

All questions carry equal weight.

CALCULATORS MAY BE USED IN THIS EXAMINATION

1. (a) Give a definition for a “communication protocol.” [2 marks]
- (b) Give three advantages of protocol layering. [3 marks]
- (c) Give two main benefits of web caching. Give an example of another application layer protocol that employs a similar caching idea. [3 marks]
- (d) Address Resolution Protocol (ARP) and Domain Name System (DNS) both depend on caches; ARP cache entry lifetimes are typically 10 minutes, while DNS cache lifetimes are on the order of days. Justify this difference. What undesirable consequences might there be in having too long a DNS cache entry lifetime? [6 marks]
- (e) Give five reasons why an application developer might choose UDP over TCP as the underlying transport protocol. [5 marks]
- (f) A high-quality version of an hour long TV programme on BBC iPlayer has a size of 350MB to download to a home computer. A home user has a broadband service with a download speed of 2Mbps. Round-trip time to the BBC iPlayer website is found to be 100ms using a ping test with default packet size of 56bytes. Assume that the access link is the bottleneck and delays other than due to propagation and transmission to be negligible.
 - Calculate the time it takes for the user to download the whole programme and the throughput of the transfer. [4 marks]
 - How much does the propagation delay matter for this transfer? [2 marks]

2. (a) Discuss five key differences between circuit switching and packet switching. [5 marks]
- (b) Datagram networks route each packet as a separate unit, independent of all others. This is not the case with virtual-circuit networks, where each data packet follows a predetermined route. Does this observation mean that virtual-circuit networks cannot route isolated packets from an arbitrary source to an arbitrary destination? Explain your answer. [3 marks]
- (c) What is CIDR (Classless InterDomain Routing)? What is the impact of using CIDR on forwarding? [4 marks]
- (d) Suppose a router has the following CIDR entries in its routing table:
- | Net/Prefix | Next Hop |
|------------------|-------------|
| 128.96.39.0/24 | Interface 0 |
| 128.96.39.128/25 | Interface 1 |
| 128.96.40.0/25 | Router 2 |
| 192.4.153.0/26 | Router 3 |
| default | Router 4 |
- For each of the following IP addresses, what does the router do if a packet with that address arrives? (a) 128.96.39.10; (b) 128.96.40.12; (c) 128.96.39.151; (d) 192.4.153.17; (e) 192.4.153.90. [5 marks]
- (e) What is the difference between routing and forwarding? [2 marks]
- (f) Compare and contrast link-state and distance-vector routing algorithms in terms of message complexity, convergence speed and robustness. [6 marks]

3. (a) Compare channel partitioning and random access MAC (medium access control) protocols at high and low loads. Which of these two classes of protocols permit statistical multiplexing and why? [7 marks]
- (b) Suppose that the data portion (including the headers) of a packet contains a sequence of 8 bytes, each corresponding to 8-bit unsigned binary representation of the following integers: 243, 81, 128, 64, 32, 16, 27, 9. Compute the Internet checksum for this data. [4 marks]
- (c) Consider the 4-bit generator, $G = 1001$, and suppose that the data portion of the bit stream to be transmitted prior to Cyclic Redundancy Check (CRC) calculation is 11001001.
- i. Show the actual bit stream transmitted. [2 marks]
- ii. Suppose the leftmost bit in the transmitted bit stream is inverted due to noise on the transmission link. Show that this error is detected at the receiver's end. [2 marks]
- (d) An upper-layer packet is split into 5 frames, each of which has an 80 percent chance of arriving undamaged. If no error control is done by the data link protocol, how many times must the message be sent on average to get the entire message through? [6 marks]
- (e) When a file is transferred between two computers, two acknowledgement strategies are possible. In the first one, the file is chopped up into packets, which are individually acknowledged by the receiver, but the file transfer as a whole is not acknowledged. In the second one, the packets are not acknowledged individually, but the entire file is acknowledged when it arrives. Discuss the relative merits of these two approaches. [4 marks]